

RESOLVED

HIGH SEVERITY

Incident Postmortem

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SERVICE	SEVERITY	OUTCOME	RECOVERY TIME
order-service	HIGH	resolved	18.0s
RETRY COUNT	LLM COST	CREATED AT	ALERT SIGNATURE
0	\$0.0000	2026-04-25	order-service:memory_leak:HIGH

Alert

order_memory_leak_bytes on order-service is 15,728,640 bytes (z-score: 3.9σ, expected: 925,214 ± 3,814,755)

Root Cause

A memory leak was intentionally injected into the order-service, causing the order_memory_leak_bytes metric to exceed the expected mean by **3.9 standard deviations**.

Fix Applied

```
restart_container({'container': 'order-service', 'reason': 'clear memory leak'})
```

Hypotheses Considered

Hypothesis 1: Memory leak intentionally injected into order-service

95% ATTEMPTED

Evidence:

- Log [2026-04-25 21:51:14]: 'FAULT: Memory leak injected. Total leak: 15.0 MB'
- Log [2026-04-25 21:51:14]: 'FAULT: Memory leak injected. Total leak: 10.0 MB'
- Metric order_memory_leak_bytes exceeded expected mean by 3.9 standard deviations

Hypothesis 2: Genuine memory leak due to a bug or inefficient code in order-service

4% NOT ATTEMPTED

Evidence:

- Metric `order_memory_leak_bytes` exceeded expected mean by 3.9 standard deviations
- Lack of similar past incidents

Hypothesis 3: Temporary memory spike due to large number of concurrent orders

1% NOT ATTEMPTED

Evidence:

- Log lines show a large number of orders created in a short period
- Metric `order_memory_leak_bytes` exceeded expected mean by 3.9 standard deviations

Audit Trail

2026-04-25T21:51:55.135273	pipeline_started
2026-04-25T21:51:55.302858	state_update
2026-04-25T21:51:55.422126	state_update
2026-04-25T21:52:04.413625	state_update
2026-04-25T21:52:05.325072	state_update
2026-04-25T21:52:11.042205	state_update
2026-04-25T21:52:11.157031	state_update
2026-04-25T21:52:11.186568	pipeline_complete